

Rishav Saigal

Senior Data Scientist | ML / GenAI Engineer | LLM Researcher & Author | Agentic AI Specialist

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SUMMARY

Senior Data Scientist and ML/GenAI Engineer with **8+ years** architecting and deploying **enterprise-grade, production-ready AI/ML systems** across healthcare, supply chain, and media verticals. Deep expertise in **Agentic AI** (Google ADK, LangGraph, CrewAI), **Retrieval-Augmented Generation (RAG)**, **Prompt Engineering**, **LLM Behavioral Research**, and **Time Series MLOps**. Author of **10 independent technical articles** (Towards AI, Towards Deep Learning, Medium, LinkedIn) plus an open, DOI-indexed **research framework & dataset** (Zenodo) on LLM behavioral drift, a self-coined **calcification effect**, autonomous ML orchestration, and AI agent production economics. Active open-source contributor (PyCaretAgent, CognitoEDA, LLM Drift Experiment, model-router). Passionate about bridging production-scale AI deployment with responsible, foundational research in multi-agent robustness and autonomous ML systems.

RESEARCH INTERESTS

LLM behavioral drift, adversarial robustness, and a self-coined calcification effect in multi-agent debate systems; autonomous ML orchestration and agentic pipeline design (AutoML, self-correcting agents); probabilistic time series forecasting with LLMs; production economics and efficiency optimisation of large-scale AI agent deployments.

EDUCATION

Indian Institute of Technology Patna	Ph.D. — Computer Science & Engineering (Pursuing)	Jul 2026 – Jun 2029 (Expected)
Birla Institute of Technology and Science, Pilani	M.Tech — Data Science & Engineering	Sep 2019 – Aug 2021
RCC Institute of Information Technology (MAKAUT)	B.Tech — Computer Science & Engineering	Aug 2013 – Jul 2017

EXPERIENCE

Cognizant

Manager (Senior Data Scientist / AI Architect)

Kolkata

Nov 2023 – Present

- **Solution Architect & Agentic AI Lead:** Spearheaded end-to-end architectural design of enterprise-grade Generative AI and Agentic AI systems on **complex GCP infrastructure (Vertex AI, Cloud Run, Gemini Enterprise)** optimised for high-quality agent delivery at scale; drove Agile sprint planning, effort estimation, and cross-functional technical delivery; led **client deliverable management** — end-to-end ownership from requirement workshops through production handoff, stakeholder-facing POC demonstrations, and sign-off governance for high-impact GenAI initiatives.
- **Team Leadership & Architectural Direction:** Led and mentored a cross-functional team of ML/GenAI engineers; defined architectural standards, technical roadmaps, and design patterns for agentic AI initiatives; drove code reviews, sprint ceremonies, and knowledge-sharing sessions ensuring consistent delivery quality across enterprise GenAI programmes.
- **Gen AI Video Generation (Google ADK, LangGraph, GCP Vertex AI, Gemini Enterprise, FastAPI):** Architected a production-ready multi-agent agentic AI pipeline on **GCP/Gemini Enterprise** ingesting meeting transcripts, applying LLM-powered NLP summarisation and storyboarding, autonomously synthesising structured sales pitch videos; integrated **token-usage observability** (cost-per-run dashboards, quality-vs-cost trade-off controls) and a **self-correction loop** (automated output validation with LLM-driven retry on failure); cut content delivery from days to **~6 hours**, reducing manual effort by **30–40%**.
- **Price Elasticity Framework (Hierarchical Linear Models, MLflow, Python, Azure ML):** Engineered a dynamic pricing solution with real-time competitor data ingestion and causal inference signals; implemented CI/CD-driven automated model retraining via MLflow; projected **10–15% revenue and market share uplift**.
- **LLM Ecosystems & RAG Applications (LangChain, FastAPI, Streamlit):** Built RAG-powered analytics chatbots enabling natural-language querying over enterprise databases with dynamic chart generation; engineered prompt-engineering QA toolchain auto-generating Gherkin/TMMi test cases, cutting manual testing effort significantly.

Genpact

Assistant Manager

Bengaluru

Jun 2022 – Nov 2023

- **Team Leadership & ML Architecture:** Managed and mentored a cross-functional team of 6 ML engineers; led Agile architecture reviews, sprint delivery, and end-to-end MLOps solution deployment; served as primary technical point-of-contact for executive stakeholder demos and quarterly business reviews.
- **Healthcare Consumer Analytics (Scikit-Learn, DataRobot, Azure ML, Python):** Designed and deployed production ML models to identify optimal account-level promotional strategies; applied causal inference, customer segmentation, and SHAP-based explainability (XAI); delivered **projected \$2.3M in cost savings at 90%+ model accuracy** across all product lines.
- **Stakeholder Delivery & Responsible AI:** Translated complex business requirements into scalable, production-ready ML specifications; conducted end-to-end solution demos and upskilled client teams on model interpretability, responsible AI governance, and proactive model monitoring.

Capgemini

Consultant

Bengaluru

Apr 2018 – Jun 2022

- **Technical Lead & Full-Lifecycle Delivery:** Led cross-functional teams of up to **15 members** across supply chain, manufacturing, and service analytics; owned end-to-end ML delivery lifecycle — from Agile requirement gathering through CI/CD-driven production deployment on AWS SageMaker and Azure ML infrastructure.
- **Demand Sensing (PySpark, TensorFlow, Keras, MLflow, AWS SageMaker):** Engineered scalable end-to-end Time Series MLOps pipeline with automated feature engineering, model selection, and continuous retraining; achieved **5% accuracy improvement, 10% better model selection**, reduced training from 3 days to 2 hours, and **60% reduction** in manual intervention.
- **Predictive Maintenance (XGBoost, Cox Proportional Hazard, Python, lifelines):** Applied time-varying survival analysis and transfer-learning-enhanced XGBoost to forecast EV component failures within defined windows; achieved **82% validation accuracy** and delivered interactive Plotly/ipywidgets visualisations for operational decision-making.
- **Additional Projects:** Service Now Analytics — LSTM/MLP deep learning models for intelligent ticket volume forecasting with custom routing algorithms (**85% validation accuracy**); Computer Vision POC — transfer-learning-based brand recognition system (OpenCV, TensorFlow) for automated image-level brand detection.

SKILLS

- **Leadership & Project Management:** Team Leadership, Agile/Scrum, Technical Roadmapping, Stakeholder Management, Effort Estimation, Architectural Design, POC Management, Impact & Risk Assessment, Cost Optimisation
- **AI/ML Development & Modeling:** Generative AI (GenAI/LLMs), RAG, Prompt Engineering, Agentic AI, TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, SciPy, Hugging Face, PyCaret, LangChain, LangGraph, CrewAI, Google ADK, OpenCV, SHAP/XAI, Transfer Learning, Survival Analysis
- **MLOps & Data Engineering:** MLflow, DataRobot, PySpark, Dask, Docker, CI/CD for ML, Model Monitoring, Feature Engineering, AWS (SageMaker, Lambda, ECS, Transcribe, Comprehend), Azure ML, GCP Vertex AI, MSSQL, MySQL, PostgreSQL, Redis
- **Programming & Visualization:** Python, JavaScript, FastAPI, Flask, REST APIs, Streamlit, DASH, Plotly, Seaborn, Matplotlib, Power BI, ipywidgets

PUBLICATIONS & INDEPENDENT RESEARCH

- The Specialized Frontier: An Inquiry into Gated AI Architectures and the Cooperative Safety Flywheel** — *Towards AI*, Aug 2026. Link: pub.towardsai.net/the-specialized-frontier-an-inquiry-into-gat...
- Claude Model Routing: Stop Scoring "Complexity." Score These Two Things Instead.** — *Towards Deep Learning*, Jul 2026. Link: www.towardsdeeplearning.com/claude-model-routing-stop-scoring-...
- LangGraph Multi-Agent Architecture: Building a Self-Critiquing AI Debate System** — *Towards AI*, May 2026. Link: pub.towardsai.net/langgraph-multi-agent-architecture-building-...
- LLM Drift Experiment: A Quantitative Framework for Measuring Behavioural Drift via Adversarial Multi-Agent Debate** — *Zenodo (Research Dataset & Framework)*, May 2026. Link: doi.org/10.5281/zenodo.20032071
- Measuring Behavioral Drift in LLMs: 22 Signals, 5 Dimensions, and the Calcification Effect** — *Towards AI*, May 2026. Link: medium.com/towards-artificial-intelligence/measuring-behaviora...
- LLM Drift Explained: Do AI Models Lose Themselves Under Adversarial Pressure?** — *Towards AI*, May 2026. Link: pub.towardsai.net/do-ai-models-lose-themselves-exploring-llm-d...
- The \$1.5 Million Difference: Why Benchmarks are Only 10% of the AI Agent Story** — *LinkedIn*, Mar 2026. Link: www.linkedin.com/pulse/15-million-difference-why-benchmarks-on...
- AutoML on Autopilot** — *Towards AI*, Mar 2026. Link: pub.towardsai.net/automl-on-autopilot-c8939bca8f8f
- CognitoEDA** — *Medium*, Aug 2025. Link: medium.com/@rishavsai/cognitoeda-bcbd3567e6d2
- Beyond Prediction: Generative AI's Probabilistic Future in Time Series** — *Medium*, Jun 2025. Link: medium.com/@rishavsai/the-generative-ai-revolution-moving-b...
- Unlocking Time Series with LLMs: A New Era with TimeCAP** — *Medium*, Jun 2025. Link: medium.com/@rishavsai/unlocking-time-series-with-llms-99a36...

PROJECTS

- Gen AI Video Generation** Jan 2026 – Present
Multi-agent pipeline (Google ADK, LangGraph, GCP Vertex AI, Gemini Enterprise) ingesting meeting transcripts, summarising and storyboarding via LLM, then synthesising sales pitch videos; architecture features a **token-cost observability layer** (per-run cost tracking, quality-vs-cost controls) and a **self-correction mechanism** (automated retry on output-validation failure).
- Media Use Case (POC)** Jun 2025 – Oct 2025
Agentic AI system (CrewAI, LangGraph, GCP) dynamically analysing ad performance, optimising budget allocation, and recommending channel-series combinations; projected **5–10% ROI uplift** and **5–7% subscription growth**.
- Price Elasticity (POC)** Jun 2025 – Jul 2025
Dynamic pricing engine using Hierarchical Linear Models (statsmodels, Python) with real-time competitor data ingestion and an automated MLflow retraining pipeline; modelled cross-product elasticities to recommend optimal price points.
- Analytics Chatbot** Jul 2024 – Dec 2025
LLM-powered chatbot (LangChain, Streamlit, PostgreSQL) enabling NL querying over enterprise databases with dynamic chart generation (Plotly) and LLM-based visual summarisation for non-technical stakeholders.
- Intelligent QA / BA Chatbot** Apr 2024 – Apr 2025
Conversational AI system (LangChain, FastAPI, Python) for internal policy Q&A, action execution, and automated test-case generation in Gherkin/TMMi formats; reduced manual QA documentation effort significantly.
- Predictive Modelling** Jul 2024 – Sep 2024
LLM-driven feature engineering pipeline converting unstructured text to numerical embeddings (OpenAI API, Scikit-Learn); incorporated SHAP explainability layer, improving model accuracy and interpretability for downstream tasks.
- LLM based Test Case Generator** Nov 2023 – Mar 2024
LLM-powered QA automation tool (LangChain, FastAPI, Python) generating test cases in Gherkin/TMMi formats from business requirements; provided intelligent, referenced Q&A, streamlining testing workflows and significantly reducing manual documentation effort.
- Healthcare Consumer Analytics** Jun 2022 – Oct 2023
Predictive promotion optimisation (Scikit-Learn, DataRobot, Azure ML) targeting account-level healthcare clients; combined customer segmentation, causal inference, and SHAP-based explainability to rank promotional strategies.
- Demand Sensing** Mar 2020 – Jun 2022
End-to-end time series MLOps pipeline (PySpark, TensorFlow, MLflow, AWS SageMaker) with automated feature engineering, model selection, and continuous retraining; designed for scalable, low-touch operation across product lines.
- Predictive Maintenance** Dec 2019 – Mar 2020
Time-varying survival analysis (Cox PH, XGBoost, lifelines, Python) forecasting EV component failures within specified windows; combined survival modelling with transfer-learning-enhanced gradient boosting and interactive Plotly visualisations.
- Service Now Analytics** Oct 2019 – Dec 2019
Ticket volume forecasting (LSTM, MLP, TensorFlow, MySQL) with custom routing algorithms for optimal agent assignment; benchmarked deep sequence models against MLP baselines with rich client reporting.
- Supply Chain Management (POC)** Apr 2019 – Oct 2019
End-to-end demand forecasting pipeline (statsmodels, NLTK Vader, pandas, Python) integrating social media sentiment as an external regressor; uni- and multi-variate models optimised via Grid Search with automated data wrangling.
- Video Analytics Solution (POC)** Apr 2018 – Mar 2019
Internal computer vision POC (OpenCV, TensorFlow, Keras, Python) for brand recognition in retail imagery; extracted and classified brand presence from individual frames to support marketing intelligence use cases.

CERTIFICATIONS

- **Generative AI Practitioner Certificate** — Dataiku Academy, Jun 2026. Link: verify.skilljar.com/c/hg2aiksrvrt2
 - **Microsoft Certified: Azure AI Fundamentals** — Microsoft, Jun 2025. Link: learn.microsoft.com/en-us/users/rishavsai/8851/credentials/...
 - **MLOps Practitioner Certificate** — Dataiku Academy, Aug 2023. Link: verify.skilljar.com/c/7djsjp74vdvj
 - **Advanced Designer Certificate** — Dataiku Academy, Jul 2023. Link: verify.skilljar.com/c/weie95jezjyz
 - **ML Practitioner Certificate** — Dataiku Academy, Jul 2023. Link: verify.skilljar.com/c/2hph8ogspvy5
 - **Core Designer Certificate** — Dataiku Academy, Jul 2023. Link: verify.skilljar.com/c/4ucupx9wj33r
- For a complete list of certifications, visit linkedin.com/in/rishav-saigal-4bb50311a.